SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Q1. Network for a Remote Rural Branch Office
Answer
The multinational company requires a network that supports cloud applications, video conferencing, secure headquarters communication, low cost, reliability, and future scalability. Since high-speed wired Internet is unavailable, a 4G/5G wireless broadband or satellite Internet connection can be selected depending on local availability.
Proposed Solution
Internet Connection:
- Primary connection: 4G/5G fixed wireless broadband where available.
- Backup: satellite Internet or a second wireless provider.
- Use automatic failover to maintain business continuity.

Networking Devices:
- Business-grade router with dual-WAN support.
- Managed Gigabit switch for wired devices.
- Wi-Fi 6 access point for wireless users.
- Firewall/security gateway for network protection.

Security:
- Site-to-site VPN between the branch and headquarters.
- Firewall with access-control rules.
- WPA3 for wireless security.
- Strong user authentication and regular firmware updates.
- Network segmentation using VLANs for employees, guests and business devices.

Constraint Analysis

Constraint	Solution
No wired high-speed Internet	4G/5G or satellite
Limited budget	Use wireless broadband with scalable equipment
Video conferencing	Prioritize real-time traffic using QoS
Uninterrupted operation	Secondary Internet connection + failover
Secure HQ communication	Site-to-site VPN
Future expansion	Managed switch and scalable router

Justification: QoS can prioritize video conferencing and business-critical cloud traffic. Dual-WAN failover provides reliability, while VPN and firewall mechanisms protect communication with headquarters. The solution can initially use a cost-effective wireless connection and later upgrade to higher-bandwidth connectivity when available.
This directly addresses the problem's bandwidth, reliability, cost and scalability constraints.
Conclusion
A 4G/5G-based dual-WAN network with VPN, firewall, QoS and Wi-Fi 6 provides a practical balance between cost, security, reliability and future expansion.

Q2. Wireless Network for 3,000+ University Students
Answer
The university needs campus-wide wireless connectivity for more than 3,000 students across classrooms, laboratories, libraries and hostels. The major constraints are large numbers of simultaneous users, limited access points, interference, security and budget.
Proposed Wireless Design
1. Access Point Placement
Access points should be strategically installed in:
- Classrooms
- Laboratories
- Libraries
- Hostels
- Common areas

Instead of placing APs only for maximum coverage, they should be positioned according to user density. High-density areas such as lecture halls and libraries should receive priority.
2. Frequency Selection
- Use 5 GHz for high-performance student devices because it provides more channels and less interference than 2.4 GHz.
- Use 2.4 GHz selectively for devices requiring longer range or compatibility.
- Where supported, use 6 GHz/Wi-Fi 6E in high-density locations.
- Reduce unnecessary transmit power to minimize overlapping coverage and interference.

3. Efficient AP Management
A centralized wireless controller can manage AP configuration, channel allocation, transmit power

and roaming.
4. Authentication and Security
Use:

- WPA3-Enterprise where supported.
- 802.1X authentication with a RADIUS server.
- Individual student credentials rather than a shared password.
- Separate guest/student/administrative networks using VLANs.

Constraint-Based Design

Constraint	Proposed Solution
3,000+ users	High-density AP placement
Limited AP budget	Install APs according to user density
Interference	Channel planning and transmit-power control
High performance	Prefer 5 GHz/6 GHz
Security	WPA3-Enterprise + 802.1X/RADIUS
Scalability	Centralized WLAN management

Conclusion
The best approach is density-based AP deployment rather than simply maximizing coverage. Strategic placement, careful frequency planning, centralized management and enterprise authentication provide good coverage and performance while controlling deployment cost.

Q3. Secure Network for a Financial Institution
Answer
The financial institution needs strong cybersecurity while allowing employees to access sensitive customer information and online banking services. The major constraints are security, performance, regulatory compliance and operational cost.
Proposed Secure Architecture
The network should follow a segmented, layered security architecture.
1. Network Segmentation
Create separate VLANs/subnets for:

- Employee systems
- Finance systems
- Customer-data systems
- Servers
- Management systems
- Guest devices

Sensitive customer-data servers should be placed in a protected server segment.
2. Firewall Deployment
Use a next-generation firewall (NGFW) at the Internet edge.
The firewall should provide:

- Access-control policies
- Application filtering
- VPN
- Intrusion prevention
- Logging and monitoring

A separate DMZ can host public-facing services so that Internet-facing systems are isolated from internal systems.
3. Authentication
Use:

- Multi-factor authentication (MFA)
- Strong passwords
- Role-based access control
- 802.1X for network access

Employees should receive only the permissions required for their jobs.
4. Intrusion Detection
Deploy IDS/IPS to monitor suspicious network traffic and detect attacks.
Security logs should be centrally collected and monitored so administrators can identify abnormal activities.
Constraint-Based Analysis

Constraint	Solution
High security	VLANs, firewall and IDS/IPS

Constraint	Solution
Sensitive customer data	Isolated server VLAN
Employee access	Role-based access control
Authentication	MFA + 802.1X
Performance	Segmentation and controlled traffic
Compliance	Logging, access control and monitoring
Budget	Use centralized security management

Conclusion

A segmented network with VLANs, NGFW, DMZ, MFA, role-based access control and IDS/IPS provides strong protection while maintaining network performance. Security controls should be deployed according to risk so that critical systems receive stronger protection without unnecessarily increasing op